École Nationale des Ponts et Chaussées

Département de première année

Bases scientifiques pour la transition énergétique 1

Séance 1 : Récapitulatif des points essentiels partie introduction

2026

Recap of what must be known for the exam

1. Noether's theorem

Energy is defined as the conserved quantity associated with symmetry by time translation.

As an example, the energy of a particle in classical mechanics is given by $E = \frac{p^2}{2m} + V$ where the first term is the kinetic energy, and the second term is the potential energy.

2. Units of energy

Energy is homogeneous to a mass times a square length divided by a square time. The SI unit of energy is the Joule : 1 J = 1 kg.m^2.s^{-2}

Some alternatives units:

- 1 eV $\approx 1.6 \times 10^{-19}$ J for atomic processes
- 1 kcal $\approx$ 4 kJ used in chemistry and biology
- 1 kW.h = 3.6 MJ for electrical energy

In spectroscopy, one can refer to energy as the corresponding frequency (ν) or wavelength (λ) of the photons having this energy: $E = h\nu = hc/\lambda$. Thus, for instance, 1 THz $\sim$ a few meV (thermal radiation in ambient conditions) and 1 µm^{-1} $\approx$ 1 eV (radiation from the sun).

For nuclear applications, one can refer to energy as the corresponding mass (m) according to the energy-mass equivalence of relativity: $E = mc^2$. The typical scale of mass energy of atoms is the GeV (about 1 GeV for a proton).

3. The usable forms of energy

Energy scales at the level of the elementary particles and processes:

radio waves (a few 10^{-7} eV): oscillation of electrons in antennas, wavelengths of a few meters.

microwaves (about 10^{-5} eV): strong enough to rotate small polar molecules (rotation of water in microwave oven), wavelength of a few cm.

thermal agitation at 300K (25 meV): thermal fluctuation of energy of magnitude $k_B T$, infrared radiation (wavelength of tens of µm), origin of the heat capacity of matter.

inter-molecular interactions (a fraction of eV): energy strong enough to trigger phase change, near infrared domain (wavelength of a few µm), origin of latent heat (evaporation, solidification).

intra-molecular interactions (a few eV): energy strong enough to break molecule and change the state of electron in atoms, origin of chemical reactions, domain of visible light and solar spectrum (wavelength of hundreds of nm).

ionization energy (hundreds of eV): energy strong enough to separate electron from nucleus, formation of plasma, UV and X-ray domain (wavelengths of a few nm),

nuclear reactions (tens or hundreds of MeV): energy involved in nuclear processes (fission, fusion), emits gamma rays

The ultimate scale of energy is given by the mass energy (GeV) that would result from complete annihilation of matter (non usable).

4. Typical orders of magnitude

Some energies representative of our everyday life:

- 1 MJ - running for 1 h
- 10 MJ - food energy of a person per day
- 200 MJ - global daily average consumption of primary energy per person

Some energies at a more global scale:

- 1 MJ: 1 ton of water released from a 100 m dam
- 10 MJ: average solar radiation per m^2 per day in France
- 1 GJ: energy of an average gasoline reservoir (50 L)
- 2 EJ (10^{18} J): annual French electricity production
- 600 EJ: global annual consumption
- 50 000 EJ: proven reserves of fossil fuels
- 5 500 000 EJ: global solar radiation on Earth per year